

Finite Element Method

Approximation of Trial Solutions for 1D Problems

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- The weak forms involve weight functions and trial solutions for the temperature, displacements, solute concentrations, and so on.
- In the FEM, weight functions and trial solutions are constructed by subdividing the domain into elements and constructing functions within each element.
- These functions have to be carefully chosen so that the FEM is **convergent**.
- The accuracy of a correctly developed FEM improves with **mesh refinement**: as element size h decreases, the solution tends to the correct solution.
- This property is of great practical importance — mesh refinement is used by practitioners to control the quality of finite element solutions.
- The accuracy of a solution is often checked by rerunning the same problem with a finer mesh. If the difference between the coarse and fine mesh solutions is small, the coarse mesh solution can be inferred to be quite accurate.

- The two necessary conditions for convergence of the FEM are **continuity** and **completeness**:

Continuity

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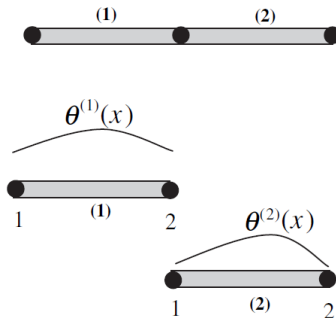
Completeness



Convergence

- Notation and nomenclature:

- $\theta(x)$: generic field variable (temperature, displacement, ...)
- $\theta^h(x)$: global FE approximation
- $\theta^e(x)$: approximation on element e
- x_1^e : x -coordinate of local node 1 of element e



A two-element mesh and the approximations in each element.

- **Continuity:** trial solutions and weight functions must be sufficiently smooth. The required smoothness depends on the order of derivatives in the weak form.
- **Completeness:** capability of a series of functions to approximate any smooth function with arbitrary accuracy. For FEM convergence it is sufficient that, as element sizes approach zero, the trial solutions and their derivatives up to and including the highest order in the weak form can assume *constant values*.
- In each element we approximate $\theta^e(x)$ by a polynomial:

$$\theta^e = \alpha_0^e + \alpha_1^e x + \alpha_2^e x^2 + \alpha_3^e x^3 + \dots$$

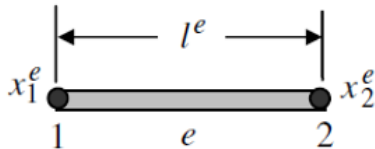
where the α_i^e are coefficients selected to satisfy continuity.

- Within each element $\theta^e(x)$ is obviously continuous. For arbitrary α_i^e , however, the approximation will *not* be continuous between elements.
- To meet C^0 continuity it is necessary that $\theta^{(1)}(x_2^{(1)}) = \theta^{(2)}(x_1^{(2)})$ (see Fig. 1).
- If coefficients α_i^e are expressed in terms of **nodal values**, it is easy to construct continuous approximations.

- The second requirement for convergence is **completeness**. Elements with a linear approximation $\theta^e = \alpha_0^e + \alpha_1^e x$ are complete: α_0^e can represent any constant function; $\alpha_1^e x$ can represent any function with a constant derivative.
- Trial solutions approximated by *incomplete* polynomials but with a complete linear part, e.g. $\theta^e = \alpha_0^e + \alpha_1^e x + \alpha_2^e x^4$, will converge, but only at the rate comparable to linear approximations.
- However, an incomplete approximation lacking the linear term, such as $\theta^e = \alpha_0^e + \alpha_2^e x^2$, **cannot** be used to develop a finite element — the resulting elements do not converge.

- Consider the simplest 1D element: two nodes, nodal values

$$\theta^e(x_1^e) \equiv \theta_1^e \quad \text{and} \quad \theta^e(x_2^e) \equiv \theta_2^e.$$



- To achieve continuity: express the approximation in terms of nodal values.
- To meet completeness: choose at least a **linear polynomial**:

$$\theta^e(x) = \alpha_0^e + \alpha_1^e x. \quad (1)$$

- Matrix form of (1):

$$\theta^e(x) = \underbrace{\begin{bmatrix} 1 & x \end{bmatrix}}_{\mathbf{p}(x)} \underbrace{\begin{bmatrix} \alpha_0^e \\ \alpha_1^e \end{bmatrix}}_{\boldsymbol{\alpha}^e} = \mathbf{p}(x) \boldsymbol{\alpha}^e \quad (2)$$

- Evaluating at the two nodes:

$$\underbrace{\begin{bmatrix} \theta_1^e \\ \theta_2^e \end{bmatrix}}_{\mathbf{d}^e} = \underbrace{\begin{bmatrix} 1 & x_1^e \\ 1 & x_2^e \end{bmatrix}}_{\mathbf{M}^e} \underbrace{\begin{bmatrix} \alpha_0^e \\ \alpha_1^e \end{bmatrix}}_{\boldsymbol{\alpha}^e} \quad (3)$$

- Inverting (3):

$$\boldsymbol{\alpha}^e = (\mathbf{M}^e)^{-1} \mathbf{d}^e \quad (4)$$

- Substituting (4) into (2):

$$\theta^e(x) = \mathbf{N}^e(x) \mathbf{d}^e, \quad \mathbf{N}^e(x) = \mathbf{p}(x)(\mathbf{M}^e)^{-1} \quad (5)$$

The row matrix $\mathbf{N}^e(x) = [N_1^e(x) \quad N_2^e(x)]$ is the *element shape function matrix*.

- From (3):

$$(\mathbf{M}^e)^{-1} = \frac{1}{l^e} \begin{bmatrix} x_2^e & -x_1^e \\ -1 & 1 \end{bmatrix}$$

where $l^e = x_2^e - x_1^e$ is the element length.

- Shape functions (using (5)):

$$\mathbf{N}^e = \frac{1}{l^e} [x_2^e - x \quad x - x_1^e] \quad (6)$$

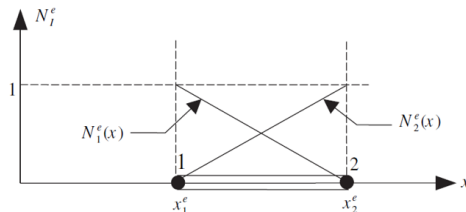


Figure 4.3 Shape functions of the two-node element.

- The shape functions satisfy the **Kronecker delta property**:

$$N_I^e(x_J^e) = \delta_{IJ}, \quad \delta_{IJ} = \begin{cases} 1 & \text{if } I = J, \\ 0 & \text{if } I \neq J. \end{cases} \quad (7)$$

- The interpolation property follows directly. Writing the approximation as

$$\theta^e(x) = \mathbf{N}^e(x) \mathbf{d}^e = \sum_{I=1}^{n_{\text{en}}} N_I^e(x) \theta_I^e,$$

and setting $x = x_J^e$:

$$\theta^e(x_J^e) = \sum_{I=1}^2 N_I^e(x_J^e) \theta_I^e = \sum_{I=1}^2 \delta_{IJ} \theta_I^e = \theta_J^e.$$

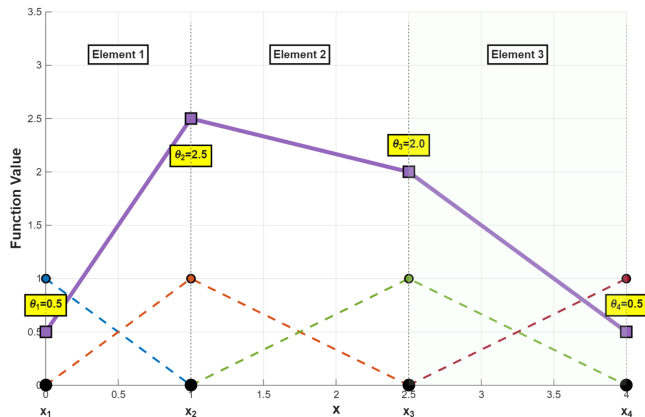
- Thus, the FE approximation is **exactly equal to the nodal values at the nodes**.
- The derivative of the trial solution is:

$$\frac{d\theta^e}{dx} = \frac{d}{dx}(\mathbf{N}^e \mathbf{d}^e) = \frac{d\mathbf{N}^e}{dx} \mathbf{d}^e = \begin{bmatrix} \frac{dN_1^e}{dx} & \frac{dN_2^e}{dx} \end{bmatrix} \begin{bmatrix} \theta_1^e \\ \theta_2^e \end{bmatrix} = \mathbf{B}^e \mathbf{d}^e, \quad \mathbf{B}^e = \frac{1}{l^e} \begin{bmatrix} -1 & +1 \end{bmatrix}. \quad (8)$$

Two-Node Linear Element (cont.)

- The trial function is the weighted sum of shape functions:

$$\theta^e(x) = \mathbf{N}^e(x)\mathbf{d}^e = \sum_{I=1}^{n_{en}} N_I^e(x) \theta_I^e.$$



Completeness

A complete polynomial expansion was used $\Rightarrow$ completeness requirement satisfied.

C^0 Continuity

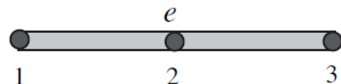
Expressing the approximation in terms of nodal values makes it straightforward to construct globally C^0 continuous functions.

- Shape functions: $\mathbf{N}^e = \frac{1}{l^e} [x_2^e - x \quad x - x_1^e]$
- Strain-displacement matrix:
 $\mathbf{B}^e = \frac{1}{l^e} [-1 \quad +1]$
- Kronecker delta property: $N_I^e(x_J^e) = \delta_{IJ}$
- Approximation: $\theta^e = \mathbf{N}^e \mathbf{d}^e, \quad \frac{d\theta^e}{dx} = \mathbf{B}^e \mathbf{d}^e$

- Start with a complete second-order polynomial:

$$\theta^e(x) = \alpha_0^e + \alpha_1^e x + \alpha_2^e x^2 = \underbrace{\begin{bmatrix} 1 & x & x^2 \end{bmatrix}}_{\mathbf{p}(x)} \underbrace{\begin{bmatrix} \alpha_0^e \\ \alpha_1^e \\ \alpha_2^e \end{bmatrix}}_{\boldsymbol{\alpha}^e} = \mathbf{p}(x) \boldsymbol{\alpha}^e \quad (9)$$

- Three coefficients $\Rightarrow$ **three nodes**.
- End nodes ensure global C^0 continuity.
- Third (mid) node placed at the element centre for symmetry; elements perform best with the midpoint node.



A three-node element.

- Nodal system $\mathbf{d}^e = \mathbf{M}^e \boldsymbol{\alpha}^e$:

$$\begin{bmatrix} \theta_1^e \\ \theta_2^e \\ \theta_3^e \end{bmatrix} = \begin{bmatrix} 1 & x_1^e & (x_1^e)^2 \\ 1 & x_2^e & (x_2^e)^2 \\ 1 & x_3^e & (x_3^e)^2 \end{bmatrix} \begin{bmatrix} \alpha_0^e \\ \alpha_1^e \\ \alpha_2^e \end{bmatrix}$$

- Combining gives $\theta^e = \mathbf{p}(\mathbf{M}^e)^{-1}\mathbf{d}^e = \mathbf{N}^e\mathbf{d}^e = \sum_{I=1}^{n_{\text{en}}} N_I^e(x) \theta_I^e$ with $n_{\text{en}} = 3$.
- The **quadratic shape functions** are:

$$\mathbf{N}^e = \frac{2}{(l^e)^2} \begin{bmatrix} (x - x_2^e)(x - x_3^e) & -2(x - x_1^e)(x - x_3^e) & (x - x_1^e)(x - x_2^e) \end{bmatrix} \quad (10)$$

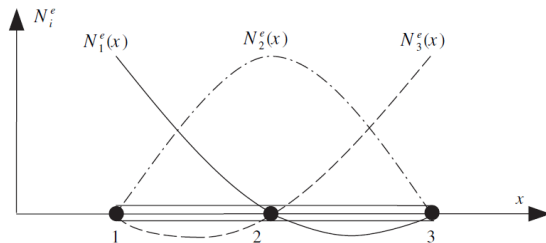


Figure 4.5 The quadratic shape functions for a three-node element.

- Each shape function satisfies the Kronecker delta property $N_I^e(x_J^e) = \delta_{IJ}$.
- Each N_I^e is nonzero only at its own node, where its value is 1.
- Within the element the shape functions are **quadratic**; the mid-node function is an upside-down parabola.

- It is not required that weight functions use the same interpolants as trial solutions.
- However, for most problems it is advantageous — and is the most common practice — to use the **same approximation** for both.
- The resulting method is called the **Galerkin FEM**, which is used throughout this course.
- Weight functions and their derivatives are then:

$$w^e(x) = \mathbf{N}^e(x) \mathbf{w}^e, \quad \frac{dw^e}{dx} = \mathbf{B}^e \mathbf{w}^e.$$

Galerkin FEM

Trial solutions and weight functions share the same shape functions: $\theta^e = \mathbf{N}^e \mathbf{d}^e$, $w^e = \mathbf{N}^e \mathbf{w}^e$, $\frac{d\theta^e}{dx} = \mathbf{B}^e \mathbf{d}^e$, $\frac{dw^e}{dx} = \mathbf{B}^e \mathbf{w}^e$.

- The global approximation gathers contributions from all n_{el} elements:

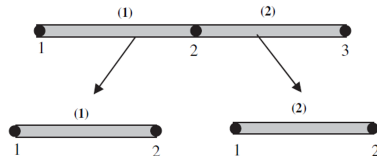
$$\theta^h = \sum_{e=1}^{n_{\text{el}}} \mathbf{N}^e \mathbf{d}^e = \left(\sum_{e=1}^{n_{\text{el}}} \mathbf{N}^e \mathbf{L}^e \right) \mathbf{d}, \quad (11)$$

$$\mathbf{w}^h = \left(\sum_{e=1}^{n_{\text{el}}} \mathbf{N}^e \mathbf{L}^e \right) \mathbf{w}, \quad (12)$$

where $\mathbf{d}^e = \mathbf{L}^e \mathbf{d}$.

- The **global shape functions**:

$$\mathbf{N} = \sum_{e=1}^{n_{\text{el}}} \mathbf{N}^e \mathbf{L}^e \quad (13)$$



Global and local node numbers for a finite element mesh.

- For the two-element mesh, the scatter matrices are:

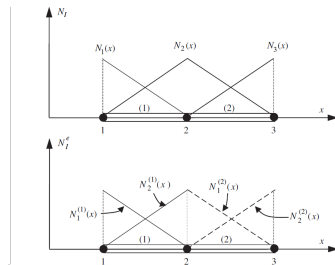
$$\mathbf{d}^{(1)} = \begin{bmatrix} \theta_1^{(1)} \\ \theta_2^{(1)} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\mathbf{L}^{(1)}} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

$$\mathbf{d}^{(2)} = \begin{bmatrix} \theta_1^{(2)} \\ \theta_2^{(2)} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{L}^{(2)}} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

- Global shape function matrix:

$$\mathbf{N} = \mathbf{N}^{(1)}\mathbf{L}^{(1)} + \mathbf{N}^{(2)}\mathbf{L}^{(2)} = \begin{bmatrix} \underbrace{N_1^{(1)}}_{N_1} & \underbrace{N_2^{(1)} + N_1^{(2)}}_{N_2} & \underbrace{N_2^{(2)}}_{N_3} \end{bmatrix}$$

- The number of global shape functions equals the number of nodes.
- Global shape functions are C^0 continuous and satisfy the Kronecker delta property.
- Any linear combination of C^0 functions is C^0 , so θ^h and w^h are guaranteed to be C^0 continuous.



Linear global (top) and element (bottom) shape functions for a two-element mesh.

- Consider a four-node cubic element with length 3, $x_1 = -1$, equally-spaced nodes:

$$x_1 = -1, x_2 = 0, x_3 = 1, x_4 = 2.$$

- Tasks: (a) construct element shape functions; (b) find the displacement field for

$$\mathbf{d}^e = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = 10^{-3} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 4 \end{bmatrix}.$$

- For a four-node (cubic) element, the **Lagrange interpolation shape functions** are:

$$N_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^4 \frac{x - x_j}{x_i - x_j} \quad (14)$$

- For $x_1 = -1$:

$$N_1(x) = -\frac{x(x-1)(x-2)}{6}$$

- For $x_2 = 0$:

$$N_2(x) = \frac{(x+1)(x-1)(x-2)}{2}$$

- For $x_3 = 1$:

$$N_3(x) = -\frac{(x+1)x(x-2)}{2}$$

- For $x_4 = 2$:

$$N_4(x) = \frac{(x+1)x(x-1)}{6}$$

- These satisfy $N_i(x_j) = \delta_{ij}$ for $i, j = 1, 2, 3, 4$.

- The displacement field is expressed as $u(x) = \sum_{i=1}^4 N_i(x) u_i$.
- Given $u_1 = 10^{-3}$, $u_2 = 0$, $u_3 = 10^{-3}$, $u_4 = 4 \times 10^{-3}$:

$$u(x) = 10^{-3} [N_1(x) + N_3(x) + 4N_4(x)]$$

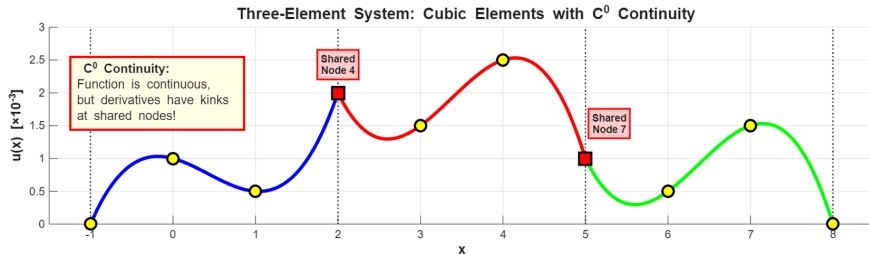
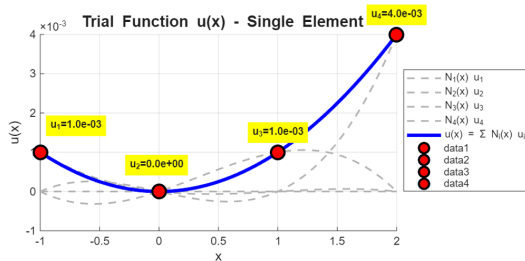
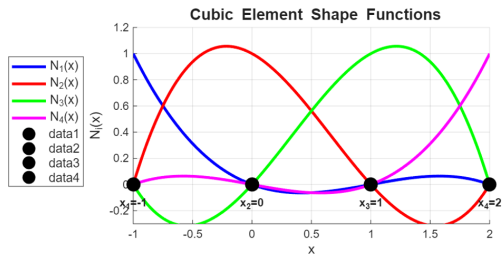
- Substituting the shape functions:

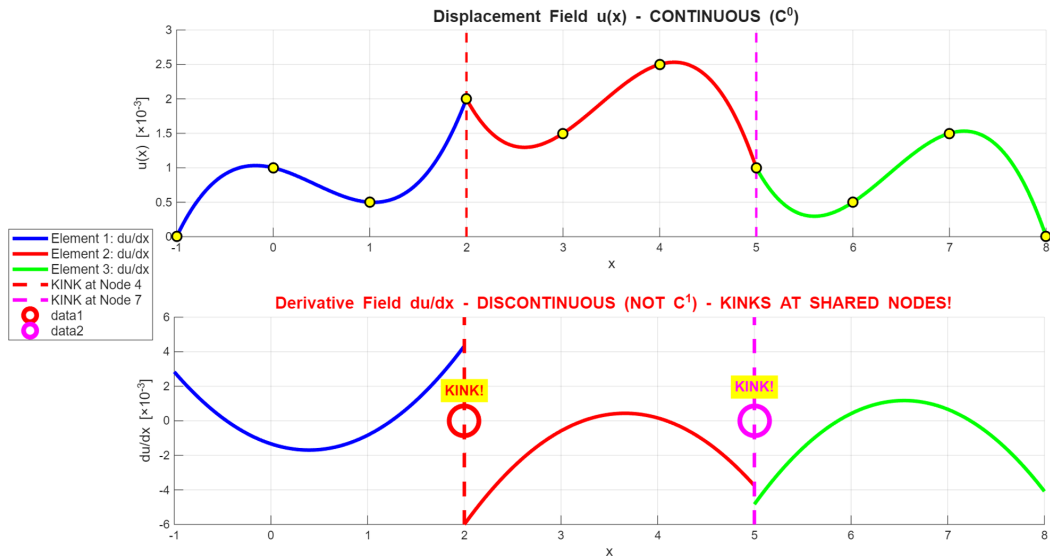
$$u(x) = 10^{-3} \left[-\frac{x(x-1)(x-2)}{6} - \frac{(x+1)x(x-2)}{2} + 4\frac{(x+1)x(x-1)}{6} \right]$$

- With a common denominator:

$$u(x) = \frac{10^{-3}}{6} \left[-x(x-1)(x-2) - 3(x+1)x(x-2) + 4(x+1)x(x-1) \right], \quad x \in [-1, 2].$$

Cubic Element (cont.)





- In general, the weak form integrals cannot be evaluated in closed form.
- **Gauss quadrature** is one of the most efficient numerical integration techniques for polynomial or near-polynomial functions.
- Consider the integral:

$$I = \int_a^b f(x) dx = ? \quad (15)$$

- Gauss formulas are defined over the **parent domain** $[-1, 1]$. Map the physical domain $[a, b]$ to $[-1, 1]$ via:

$$x = \frac{1}{2}(a+b) + \frac{1}{2}\xi(b-a) \quad (16)$$

or equivalently using linear shape functions:

$$x = x_1 N_1(\xi) + x_2 N_2(\xi) = a \frac{1-\xi}{2} + b \frac{\xi+1}{2} \quad (17)$$



Mapping of the one-dimensional domain from the parent domain $[-1, 1]$ to the physical domain $[a, b]$.

- The differential transforms as:

$$dx = \frac{1}{2}(b-a) d\xi = \frac{l}{2} d\xi = J d\xi \quad (18)$$

where $J = (b-a)/2$ is the **Jacobian**.

- The integral becomes:

$$I = J \int_{-1}^1 f(\xi) d\xi = J\hat{I}, \quad \hat{I} = \int_{-1}^1 f(\xi) d\xi.$$

- Gauss quadrature approximation:**

$$\hat{I} \approx W_1 f(\xi_1) + W_2 f(\xi_2) + \cdots = \underbrace{[W_1 \quad W_2 \quad \cdots \quad W_n]}_{\mathbf{W}^T} \underbrace{\begin{bmatrix} f(\xi_1) \\ f(\xi_2) \\ \vdots \\ f(\xi_n) \end{bmatrix}}_{\mathbf{f}} = \mathbf{W}^T \mathbf{f} \quad (19)$$

- The number of integration points n_{gp} needed to integrate a polynomial of order p *exactly*:

$$n_{\text{gp}} \geq \frac{p+1}{2}$$

- The Gauss quadrature points and weights are tabulated below.

n_{gp}	Location, ξ_i	Weights, W_i
1	0.0	2.0
2	$\pm 1/\sqrt{3} = \pm 0.5773502692$	1.0
3	± 0.7745966692	0.555 555 5556
	0.0	0.888 888 8889
4	± 0.8611363116	0.347 854 8451
	± 0.3399810436	0.652 145 1549
	± 0.9061798459	0.236 926 8851
5	± 0.5384693101	0.478 628 6705
	0.0	0.568 888 8889
	± 0.9324695142	0.171 324 4924
6	± 0.6612093865	0.360 761 5730
	± 0.2386191861	0.467 913 9346

Problem

Evaluate $I = \int_2^5 (x^3 + x^2) dx$ using two-point Gauss quadrature.

- The integrand is degree 3, so $2n_{\text{gp}} - 1 = 3 \Rightarrow n_{\text{gp}} = 2$.
- From the table: $W_1 = W_2 = 1$, $\xi_1 = -1/\sqrt{3}$, $\xi_2 = 1/\sqrt{3}$.
- Map $[2, 5] \rightarrow [-1, 1]$ with $a = 2$, $b = 5$:

$$x = \frac{1}{2}(a + b) + \frac{1}{2}\xi(b - a) = 3.5 + 1.5\xi, \quad J = \frac{b-a}{2} = 1.5.$$

$$f(\xi) = (3.5 + 1.5\xi)^3 + (3.5 + 1.5\xi)^2.$$

- Applying the quadrature formula:

$$I = J[W_1 f(\xi_1) + W_2 f(\xi_2)] = 1.5[f(-1/\sqrt{3}) + f(1/\sqrt{3})] = 37.818 + 153.432 = \mathbf{191.25}.$$

- Verification by exact integration:

$$\int_2^5 (x^3 + x^2) dx = \left[\frac{x^4}{4} + \frac{x^3}{3} \right]_2^5 = 197.917 - 6.667 = \mathbf{191.25} \checkmark$$

Next: Finite Element Approximation — 1D Problems I

Thanks for your attention!